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United States Patent	12398661
Kind Code	B2
Date of Patent	August 26, 2025
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Containment ring for gas turbine engine

Abstract

A containment ring assembly for a turbine casing assembly, including: a first inner containment ring; a second inner containment ring; an outer containment ring, the first inner containment ring and the second inner containment ring being radially inward from the outer containment ring, wherein a radial space is located between the outer containment ring and the first inner containment ring and the second inner containment ring, the radial space allowing deformation of the first inner containment ring and the second inner containment ring.

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Appl. No.: **18/426019**

Filed: **January 29, 2024**

Prior Publication Data

Document Identifier	Publication Date
US 20250243785 A1	Jul. 31, 2025

Publication Classification

Int. Cl.: **F01D25/24** (20060101); **F02C7/00** (20060101)

U.S. Cl.:

CPC **F01D25/243** (20130101); **F02C7/00** (20130101);

Field of Classification Search

CPC: F01D (21/045)

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Background/Summary

BACKGROUND
(1) This disclosure relates to structures for use in gas turbine engines, and more particularly to a containment ring for use in a gas turbine engine.

(2) Gas turbine engines have a turbine casing assembly. The turbine casing assembly may include a containment ring. It is desirable to decrease the weight of a containment ring while providing the desired containment.

BRIEF DESCRIPTION

(3) Disclosed is a containment ring assembly for a turbine casing assembly, including: a first inner containment ring; a second inner containment ring; an outer containment ring, the first inner containment ring and the second inner containment ring being radially inward from the outer containment ring, wherein a radial space is located between the outer containment ring and the first inner containment ring and the second inner containment ring, the radial space allowing deformation of the first inner containment ring and the second inner containment ring.

(4) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the first inner containment ring is secured to the second inner containment ring by a plurality of frangible tab members.

(5) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the outer containment ring is secured to the first inner containment ring by a plurality of frangible tab members.

(6) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the outer containment ring is secured to the second inner containment ring by a plurality of frangible tab members.

(7) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the second inner containment ring is secured to a flange portion by a plurality of frangible tab members.

(8) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the plurality of frangible tab members are separated by a plurality of openings.

(9) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the first inner containment ring defines a first stage containment zone and the outer containment ring defines a second stage containment zone and the second inner containment ring defines a third stage containment zone.

(10) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the first inner containment ring is secured to the second inner containment ring by a plurality of frangible tab members.

(11) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the outer containment ring is secured to the first inner containment ring by a plurality of frangible tab members.

(12) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the outer containment ring is secured to the second inner containment ring by a plurality of frangible tab members.

(13) Also disclosed is a turbine casing assembly, including: an outer structural case; and a containment ring, including: a first inner containment ring; a second inner containment ring; an outer containment ring, the first inner containment ring and the second inner containment ring being radially inward from the outer containment ring, wherein a radial space is located between the outer containment ring and the first inner containment ring and the second inner containment ring, the radial space allowing deformation of the first inner containment ring and the second inner containment ring.

(14) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the first inner containment ring is secured to the second inner containment ring by a plurality of frangible tab members.

(15) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the outer containment ring is secured to the first inner containment ring by

a plurality of frangible tab members.
(16) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the outer containment ring is secured to the second inner containment ring by a plurality of frangible tab members.
(17) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the second inner containment ring is secured to a flange portion by a plurality of frangible tab members and the flange portion is secured to the outer structural case.
(18) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the plurality of frangible tab members are separated by a plurality of openings.
(19) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the first inner containment ring defines a first stage containment zone and the outer containment ring defines a second stage containment zone and the second inner containment ring defines a third stage containment zone.
(20) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the first inner containment ring is secured to the second inner containment ring by a plurality of frangible tab members.
(21) In addition to one or more of the features described above, or as an alternative to any of the foregoing embodiments, the outer containment ring is secured to the first inner containment ring by a plurality of frangible tab members.
(22) Also disclosed is a gas turbine engine, including: a compressor section; a combustor; a turbine section; and a turbine casing assembly, including: an outer structural case; and a containment ring, including: a first inner containment ring; a second inner containment ring; an outer containment ring, the first inner containment ring and the second inner containment ring being radially inward from the outer containment ring, wherein a radial space is located between the outer containment ring and the first inner containment ring and the second inner containment ring, the radial space allowing deformation of the first inner containment ring and the second inner containment ring.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:
(2) FIG. **1** is a schematic cross-sectional view of a gas turbine engine;
(3) FIG. **2** is a perspective view of a containment ring and rotors in accordance with the present disclosure;
(4) FIG. **3** is a perspective cross-sectional view of the containment ring and rotors illustrated in FIG. **2**;
(5) FIG. **4** is an enlarged perspective cross-sectional view of a portion of the containment ring in accordance with the present disclosure; and
(6) FIG. **5** is an illustration of containment of a rotor by the containment ring in accordance with the present disclosure.
DETAILED DESCRIPTION
(7) A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the FIGS.
(8) FIG. **1** illustrates a gas turbine engine **10**, generally comprising in serial flow communication a compressor section **14** for pressurizing the air, a combustor section **16** in which the compressed air is mixed with fuel and ignited for generating an annular stream of hot combustion gases, and a turbine section **18** for extracting energy from the combustion gases. Some of the rotatable

components of the gas turbine engine **10** rotate about a longitudinal center axis **20** of the gas turbine engine **10**.
(9) The gas turbine engine **10** has a "cold" section **22** and a "hot" section **24**. The cold section **22** includes those components of the gas turbine engine **10** which are upstream (relative to the direction gases flow through the gas turbine engine **10**) of the combustor **16** and have thus not been exposed to the hot combustion gases. The hot section **24** includes the combustor **16** and those components of the gas turbine engine **10** which are downstream of the combustor **16**. The components of the hot section **24** are thus exposed to the hot combustion gases generated in the combustor **16**. The gases flowing through the cold section **22** have a lower temperature than the gases flowing through the hot section **24**.
(10) The hot section **24** includes the combustor **16**, the turbine section **18** and a case downstream of the turbine section **18** for conveying the exhaust gases. The turbine section **18** includes one or more rotors **26** each having a plurality of rotor blades **28** secured to a hub **29** which rotates about the center axis **20** and extract energy from the combustion gases. In one implementation the turbine blades **28** are integrally formed with the hub **29** in order to form a single component referred to as an integrally bladed rotor or IBR **26**. The hot section **24** includes stationary bodies which enclose other components of the hot section **24** and define the gas path for the hot combustion gases. These stationary bodies are sometimes referred to as casings or cases which collectively define radially-outer boundaries of the gas turbine engine.
(11) Referring now to at least FIGS. **1-4**, the casing of the gas turbine engine **10** includes a turbine casing assembly **30** which is part of the hot section **24**. The turbine casing assembly **30** is a group of casing components that form part of the turbine section **18** and enclose the combustion gases. The turbine casing assembly **30** may be provided as disassembled components which may then be assembled in a suitable facility. The turbine casing assembly **30** includes an outer structural case **32**, a containment ring or containment ring assembly **34**, and at least one vane ring **36**. The at least one vane ring **36** having at least one vane **38** secured thereto. In one embodiment, the least one vane ring **36** having the at least one vane **38** may be formed as a single component.
(12) In one embodiment, the outer structural case **32**, the containment ring or containment ring assembly **34**, and the least one vane ring **36** have circular portions or configurations. As illustrated in the attached FIGS., a three stage rotor assembly is illustrated or in other words three rotors **26** are illustrated and may be referred to as a first stage rotor **26**′, a second stage rotor **26**″ and a third stage rotor **26**‴ respectively.
(13) The present disclosure is generally directed to a containment ring or containment ring assembly **34** having a three containment ring configuration where each ring has brittle joints. The purpose of these brittle joints is to fracture to ensure that each containment ring of the containment ring assembly **34** can absorb the kinetic energy of a fragment and deform freely without transmitting the load to the engine structure in the event of an overspeed tri-hub failure. The geometry of the containment ring or containment ring assembly **34** allows it to meet containment criteria even in an engine with a compact profile having three narrow turbine rotors.
(14) FIG. **2** is a perspective view of a containment ring or containment ring assembly **34** configured to surround three rotors **26** in accordance with the present disclosure and FIG. **3** is a cross-sectional view of the containment ring or containment ring assembly **34** illustrated in FIGS. **1** and **2**. FIG. **4** is an enlarged perspective cross-sectional view of a portion of the containment ring or containment ring assembly **34** in accordance with the present disclosure.
(15) In accordance with the present disclosure, the containment ring or containment ring assembly **34** has three radially superposed containment zones, defined by a first stage containment zone **40** comprising a first inner containment ring **42**, and a second stage containment zone **44** comprising an outer containment ring **46** and a third stage containment zone **48** comprising a second inner containment ring **50**.
(16) The first inner containment ring **42** being aligned with the first stage rotor **26**′ and the outer

containment ring **46** being aligned with the second stage rotor **26″** and the second inner containment ring **50** being aligned with the third stage rotor **26‴**. The containment rings **42**, **46**, and **50** are configured and dimensioned to absorb the kinetic energy of a fragment by plastic deformation of a respective one of the first stage rotor **26′**, the second stage rotor **26″**, and the third stage rotor **26‴**.

(17) The outer containment ring **46** may also be referred to as a central containment ring superimposed on the two inner rings **42** and **50** and is configured to overlap the two lower inner rings **42** and **50** and meet a critical angle required to contain the fragments from a rotor stage failure.

(18) A radial space **52** is located between the two containment rings **42**, **50** and the containment ring **46**. The radial space **52** allows the first inner containment ring **42** and the second inner containment ring **50** to deform freely with regard to the outer containment ring **46**. In other words, the radial space **52** allows deformation of the first inner containment ring **42** and the second inner containment ring **50** during a containment event (e.g., the containment rings or containment ring assembly **34** retaining or containing a fragment). In accordance with the present disclosure, the thickness of the containment rings **42**, **46** and **50** can be adapted depending upon the rotor **26** geometries and speed.

(19) Referring now to at least FIGS. **1-4**, the first inner containment ring **42** has a main body portion **54**. The main body portion **54** has a forward end **56** and an aft end **58**, an outer periphery **60**, and an inner periphery **62**. As used herein, the outer periphery **60** is radially outward from the inner periphery **62**. In addition, and when the inner containment ring **42** is installed in the engine **10**, the outer periphery **60** is radially further from the central axis **20** than the inner periphery **62**. In addition and as used herein, the forward end **56** is closer to the combustor section **16** than the aft end **58** when the containment ring or containment ring assembly **34** is secured to the engine **10**.

(20) The second inner containment ring **50** has a main body portion **70**. The main body portion **70** has a forward end **72** and an aft end **74**, an outer periphery **76**, and an inner periphery **78**. As used herein, the outer periphery **76** is radially outward from the inner periphery **78**. In addition, and when the second inner containment ring **50** is installed in the engine **10**, the outer periphery **76** is radially further from the central axis **20** than the inner periphery **78**. In addition and as used herein, the forward end **72** is closer to the combustor section **16** than the aft end **74** when the containment ring or containment ring assembly **34** is secured to the engine **10**.

(21) The outer containment ring **46** has a main body portion **80**. The main body portion **80** has a forward end **82** and an aft end **84**, an outer periphery **86**, and an inner periphery **88**. As used herein, the outer periphery **86** is radially outward from the inner periphery **88**. In addition, and when the outer containment ring **46** is installed in the engine **10**, the outer periphery **86** is radially further from the central axis **20** than the inner periphery **88**. In addition and as used herein, the forward end **82** is closer to the combustor section **16** than the aft end **84** when the containment ring or containment ring assembly **34** is secured to the engine **10**.

(22) The second inner containment ring **50** also has a flange portion **90** secured thereto. The flange portion **90** is configured to secure the second inner containment ring **50** and the containment ring or containment ring assembly **34** to the outer structural case **32**. The flange portion **90** being secured to the main body portion **70** via a plurality of tabs or frangible tab members **92**. In one embodiment, each of the plurality of tabs or frangible tabs members **92** are separated from each other by a plurality of openings **94**. Still further and in one non-limiting embodiment, the tab members **94** are integrally formed with the second inner containment ring **50** so that they form a single unitary structure.

(23) In addition, the aft end **58** of the main body portion **54** of the first inner containment ring **42** is secured to the forward end **72** of the main body portion **70** of the second inner containment ring **50** by a plurality of tabs or frangible tab members **96**. In one embodiment, each of the plurality of tabs or frangible tab members **96** are separated from each other by a plurality of openings **98**. In one

non-limiting embodiment, the first inner containment ring **42**, the second inner containment ring **50** and the flange portion **90** are integrally formed so that they form a single unitary structure.
(24) The forward end **82** of the main body portion **80** of the outer containment ring **46** is secured to the main body portion **54** of the first inner containment ring **42** by a plurality tabs or frangible tab members **100**. In one embodiment, each of the plurality of tabs or frangible tab members **100** are separated from each other by a plurality of openings **102**.
(25) The aft end **84** of the main body portion **80** of the outer containment ring **46** is secured to the flange portion **90** by a plurality tabs or frangible tab members **104**. In one embodiment, each of the plurality of tabs or frangible tab members **104** are separated from each other by a plurality of openings **106**.
(26) In one embodiment, the radial space **52** is in fluid communication with the plurality of openings **94**, **98**, **102** and **106**. In addition, the plurality of tabs or frangible tab members **92**, **96**, **100** and **104** provide brittle frangible zones or joints that can fracture to ensure that each containment ring of the containment ring assembly **34** can absorb the kinetic energy of a fragment and deform freely without transmitting the load to the engine structure in the event of an overspeed tri-hub failure.
(27) In one non-limiting embodiment, the aft end **84** is secured to the flange portion **90** and the forward end **82** is secured to the first inner containment ring **42**. In one non-limiting embodiment, the securement of the respective ends **82** and **84** is achieved by rivets **89** or any other suitable means e.g. welding or equivalents thereof provide to a tight fit. These connections or tight fits will improve the dynamic response of the inner containment rings **42** and **50** with respect to the outer containment ring **46**. In an alternative embodiment, the flange portion **90** is integrally formed with the outer containment ring **46** as opposed to the second inner containment ring **50** and the aft end **74** of the main body portion **70** of the second inner containment ring **50** is secured to the flange portion **90**. In one non-limiting embodiment, the first inner containment ring **42**, the second inner containment ring **50**, the outer containment ring **46** and the flange portion **90** may be formed from nickel based alloys or any other suitable material.
(28) Referring now to FIGS. **1-5**, the first inner containment ring **42**, the second inner containment ring **50** and the outer containment ring **46** are designed to absorb the kinetic energy of disc fragments **108** in the direction of arrows **110** in the event of a tri-hub fracture event. For example, in the event the first stage rotor **26′** and/or the second stage rotor **26″** and/or the third stage rotor **26′″** has separated into fragments **108** illustrated in FIG. **5**. Note, the description of a tri-hub fracture event is merely provided as an example and various embodiments of the present disclosure are not limited to this specific event. FIG. **5** is an illustration of containment of portions or fragments of anyone of the rotors **26′**, **26″** and **26′″** by the containment ring or containment ring assembly **34** in accordance with the present disclosure.
(29) For example and in an uncontrolled overspeed event, rotating parts (turbine discs, compressor rotor, impeller, etc.) can fracture in three parts (tri-hub) and the high energy fragments are contained by the containment ring or containment ring assembly **34**. As such and in accordance with the present disclosure, these fragments **108** are contained using the containment ring or containment ring assembly **34** of the present disclosure.
(30) The present disclosure is directed to a containment ring or containment ring assembly **34** having a three containment ring configuration where each ring has brittle joints. The purpose of these brittle joints is to fracture to ensure that each containment ring of the containment ring assembly **34** can absorb the kinetic energy of a fragment and deform freely without transmitting the load to the engine structure in the event of an overspeed tri-hub failure. The geometry of the containment ring or containment ring assembly **34** allows it to meet containment criteria even in an engine with a compact profile having three narrow turbine rotors.
(31) As such, an architecture of superimposed containment rings **42**, **46** and **50** is provided to enable the design of brittle joints for each containment ring **42**, **46** and **50**.

(32) The outer containment ring **46** is configured to stiffen the containment assembly **34** by means of two interference joints arranged at each end of the two rings, thus avoiding the problem of vibration generated by the introduction of fragile joints.
(33) The outer containment ring **46** is superimposed on the two inner rings **42** and **50** and the outer containment ring **46** is configured to overlap the two lower rings **42** and **50** in order to a critical angle required to contain the fragments **108**.
(34) In an uncontrolled overspeed event, the critical rotating parts (turbine discs, compressor rotor, impeller, etc.) can fracture in three parts (tri-hub) and the high energy fragments need to be radially contained. When one of these fragment **108** hits the containment ring or containment assembly **34**, the kinetic energy (KE) is absorbed by the potential energy (PE) of the rings **42**, **46** and **50**, which deform under the load. However, to allow deformation of the individual rings **42**, **46** and **50**, it is necessary for each of the containment rings **42**, **46** and **50** to be supported by a fragile joint designed to break on contact with the fragment **108**.
(35) The necessary design requirement for a containment ring or containment ring assembly **34** is to maintain the integrity of the outer diameter of the ring, i.e. to prevent the fragment from piercing the whole radial thickness. However, and in order to meet this requirement, it requires large axial space to design the containment rings. The present disclosure comprises superimposing the outer containment ring **46** on top of the two inner rings **42**, **50**, thus meeting the minimum area required to contain fragments of each the rotors **26′**, **26″** and **26′″**.
(36) The advantage of this concept allows the gas turbine engine **10** to be designed in a compact axial space while maintaining the containment requirements. As such, this allows the containment ring or containment assembly **34** to be used in a very small engine centerline. This allows the weight of the engine **10** to be within desired ranges without lengthening of the containment area, and therefore the overall length of the engine **10**.
(37) The term “about” is intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application. For example, “about” can include a range of ±8% or 5%, or 2% of a given value.
(38) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, element components, and/or groups thereof.
(39) While the present disclosure has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the present disclosure. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this present disclosure, but that the present disclosure will include all embodiments falling within the scope of the claims.

Claims

1. A containment ring assembly for a turbine casing assembly, comprising: a first inner containment ring, the first inner containment ring having a thickened main body portion, the thickened main body portion having a forward end and an aft end; a second inner containment ring, the second inner containment ring having a thickened main body portion, the thickened main body portion having a forward end and an aft end; an outer containment ring, the outer containment ring having

a thickened main body portion, the thickened main body portion having a forward end and an aft end, the thickened main body portion of the outer containment ring being at least partially located between the aft end of the thickened main body portion of the first inner containment ring and the forward end of the thickened main body portion of the second inner containment ring, the first inner containment ring and the second inner containment ring being radially inward from the outer containment ring, wherein a radial space is located between the outer containment ring and the first inner containment ring and the second inner containment ring, the radial space allowing the first inner containment ring and the second inner containment ring to deform freely with regard to the outer containment ring.
2. The containment ring assembly as in claim 1, wherein the first inner containment ring is secured to the second inner containment ring by a first plurality of frangible tab members.
3. The containment ring assembly as in claim 2, wherein the outer containment ring is secured to the first inner containment ring by a second plurality of frangible tab members.
4. The containment ring assembly as in claim 3, wherein the outer containment ring is secured to the second inner containment ring by a third plurality of frangible tab members.
5. The containment ring assembly as in claim 4, wherein the second inner containment ring is secured to a flange portion by a fourth plurality of frangible tab members.
6. The containment ring assembly as in claim 5, wherein each of the first, second, third and fourth plurality of frangible tab members are separated from each other by one of a plurality of openings.
7. The containment ring assembly as in claim 1, wherein the first inner containment ring defines a first stage containment zone and the outer containment ring defines a second stage containment zone and the second inner containment ring defines a third stage containment zone.
8. The containment ring assembly as in claim 7, wherein the first inner containment ring is secured to the second inner containment ring by a first plurality of frangible tab members.
9. The containment ring assembly as in claim 8, wherein the outer containment ring is secured to the first inner containment ring by a second plurality of frangible tab members.
10. The containment ring assembly as in claim 9, wherein the outer containment ring is secured to the second inner containment ring by a third plurality of frangible tab members.
11. A turbine casing assembly, comprising: an outer structural case; and a containment ring, comprising: a first inner containment ring, the first inner containment ring having a thickened main body portion, the thickened main body portion having a forward end and an aft end; a second inner containment ring, the second inner containment ring having a thickened main body portion, the thickened main body portion having a forward end and an aft end; an outer containment ring, the outer containment ring having a thickened main body portion, the thickened main body portion having a forward end and an aft end, the thickened main body portion of the outer containment ring being at least partially located between the aft end of the thickened main body portion of the first inner containment ring and the forward end of the thickened main body portion of the second inner containment ring, the first inner containment ring and the second inner containment ring being radially inward from the outer containment ring, wherein a radial space is located between the outer containment ring and the first inner containment ring and the second inner containment ring, the radial space allowing deformation of the first inner containment ring and the second inner containment ring to deform freely with regard to the outer containment ring.
12. The turbine casing assembly as in claim 11, wherein the first inner containment ring is secured to the second inner containment ring by a first plurality of frangible tab members.
13. The turbine casing assembly as in claim 12, wherein the outer containment ring is secured to the first inner containment ring by a second plurality of frangible tab members.
14. The turbine casing assembly as in claim 13, wherein the outer containment ring is secured to the second inner containment ring by a third plurality of frangible tab members.
15. The turbine casing assembly as in claim 14, wherein the second inner containment ring is secured to a flange portion by a fourth plurality of frangible tab members and the flange portion is

secured to the outer structural case.
16. The turbine casing assembly as in claim 15, wherein each of the first, second, third and fourth plurality of frangible tab members are separated from each other by one of a plurality of openings.
17. The turbine casing assembly as in claim 11, wherein the first inner containment ring defines a first stage containment zone and the outer containment ring defines a second stage containment zone and the second inner containment ring defines a third stage containment zone.
18. The turbine casing assembly as in claim 17, wherein the first inner containment ring is secured to the second inner containment ring by a first plurality of frangible tab members.
19. The turbine casing assembly as in claim 18, wherein the outer containment ring is secured to the first inner containment ring by a second plurality of frangible tab members.
20. A gas turbine engine, comprising: a compressor section; a combustor; a turbine section; and a turbine casing assembly, comprising: an outer structural case; and a containment ring, comprising: a first inner containment ring, the first inner containment ring having a thickened main body portion, the thickened main body portion having a forward end and an aft end; a second inner containment ring, the second inner containment ring having a thickened main body portion, the thickened main body portion having a forward end and an aft end; an outer containment ring, the outer containment ring having a thickened main body portion, the thickened main body portion having a forward end and an aft end, the thickened main body portion of the outer containment ring being at least partially located between the aft end of the thickened main body portion of the first inner containment ring and the forward end of the thickened main body portion of the second inner containment ring, the first inner containment ring and the second inner containment ring being radially inward from the outer containment ring, wherein a radial space is located between the outer containment ring and the first inner containment ring and the second inner containment ring, the radial space allowing deformation of the first inner containment ring and the second inner containment ring to deform freely with regard to the outer containment ring, the first inner containment ring being aligned with a first stage rotor of the turbine section, the outer containment ring being aligned with a second stage rotor of the turbine section, and the second inner containment ring being aligned with a third stage rotor of the turbine section.